

Call & Response MIDI — User Manual (v1.0.5)

Viral Samples | Intelligent Algorithmic Phrase Generator & 303-Style Sequencer

Formats: VST3, Audio Unit (AU), Standalone | Platforms: macOS (Apple Silicon & Intel) & Windows (64-bit)

1. Overview & Architecture

Call & Response MIDI is a dual-lane algorithmic melody generator, performance sequencer, and MIDI composition instrument inspired by classic Roland TB-303 phrasing and the universal musical conversation of *Call & Response*.

Instead of random notes, Call & Response MIDI generates pairs of musical phrases that dialogue with one another:

- **Call Lane:** The question, theme, or lead hook.
- **Response Lane:** The answer, variation, or harmonic resolution shaped by sophisticated modal and rhythmic response strategies.

A built-in analog-modelled 303 preview synth with resonant low-pass filter allows instant standalone auditioning without having to wire a virtual instrument first.

2. System Requirements & Compatibility

2.1 macOS

- **Operating System:** macOS 11.0 (Big Sur) or higher (Monterey, Ventura, Sonoma, Sequoia fully supported).
- **CPU Architectures:**
- **Apple Silicon** (`arm64`): M1, M2, M3, M4, Pro, Max, Ultra native).
- **Intel Macs** (`x86_64`): Core i5, i7, i9, Xeon native).
- **Plugin Formats:** VST3 (64-bit universal), Audio Unit / AU (64-bit universal), Standalone Application.

2.2 Windows

- **Operating System:** Windows 10 or Windows 11 (64-bit).
- **CPU Architectures:** 64-bit x86-64 (Intel Core / AMD Ryzen).
- **Plugin Formats:** VST3 (64-bit), Standalone Application (`.exe`).

2.3 DAW Compatibility

Ableton Live 10/11/12 (macOS & Windows), Logic Pro X (macOS), FL Studio 20/21/24 (macOS & Windows), Cubase, Studio One, Reaper, Bitwig Studio, Pro Tools (via AU/VST3 wrapper).

3. Header & Utility Controls

[CALL & RESPONSE] VIRAL SAMPLES | v1.0.5 [STEPS] [PIANO ROLL] [SPLIT] [A] [B] [C] [D] [E]

3.1 Preset System (Slots A–H)

The plugin includes 8 rapid-access preset memory slots located in the top header:

- **Save Current State:** **Cmd+Click** (or **Shift+Click**, **Alt+Click**, **Ctrl+Click**) on any slot button (**A** through **H**). The status bar confirms: "Saved current state to Preset Slot X".
- **Load Preset:** **Double-Click** on any slot button to immediately recall its pattern, scale, key, and settings.
- **Context Menu:** **Right-Click** on any slot button to access the popup menu:
 - *Save Current State to Slot X*
 - *Load Preset Slot X*
- **Visual Status:** Occupied slots feature a bright cyan indicator dot and highlighted border. Single-clicking inspects slot occupancy without overwriting unsaved work.

3.2 Undo & Redo History

- **Undo / Redo Buttons:** Store up to **24 states** of musical edits, generations, transpositions, and mutations. If an edit or random roll doesn't fit your arrangement, click **Undo** to instantly restore your previous pattern.

3.3 Audio Preview & Panic

- **PREVIEW: ON / MUTED:** Toggles the internal 303 synthesizer engine. Mute this button when routing MIDI output directly to external hardware or software synthesizers.
- **PANIC:** Immediately issues *All-Notes-Off* and *All-Sound-Off* commands across all 16 MIDI channels to eliminate hanging notes.

4. Generative Engine & Controls

4.1 Musical Controls

- **Key:** Sets the harmonic root (**C** through **B**).
- **Scale:** Choose from **35 curated musical scales**:
 - *Diatonic & Modes:* Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Locrian, Lydian Dominant, Aeolian b5, Dorian b2, Mixolydian b6.
 - *Pentatonic & Blues:* Minor Pentatonic, Major Pentatonic, Blues Minor, Blues Major, Egyptian Pentatonic.
 - *Exotic & Harmonic:* Phrygian Dominant, Double Harmonic, Hungarian Minor, Romanian Minor, Whole Tone, Diminished, Byzantine, Persian, Arabic, Hirajoshi, In Sen, Kumoi, Pelog.
 - *Minimalist & Arp Pools:* Power Fifths, Root + b7, Minor 7 Arp, Dominant 7 Arp, Diminished 7 Arp.
- **Root Octave:** Sets baseline octave (**C1** through **C5**).

- **Steps:** Select loop lengths of **4, 8, 12, or 16 steps**.

4.2 Style & Response Strategy

- **Style (40 Styles):** Governs rhythmic bias, syncopation, rest likelihood, and slide/accent characteristics (e.g., *Classic Acid, Raw Acid, Squelchy Acid, Dark Rave, Driving Techno, Minimal Techno, Jackin House, Deep House, Hooverish Riff, Breaky Electro*).
- **Response Strategy (38 Types):** Dictates how the response answers the call:
 - *Same rhythm + resolve, Invert contour, Answer downward, Mirror contour, Echo, Response resolves to root, Tighter answer, Sparser answer, Transpose motif up/down.*
- **Motif Type (24 Types):** Shapes the opening lead-in hook (*Pulse, Offbeat, Syncopated, Push, Rolling, Acid nibble, Arp fragment, Techno cell, House bounce*).
- **Ending Behaviour & Phrase Mode:** Configures cadence resolution and relationship between Call and Response phases (*Call then response, Call response overlapped, Call solo*).

4.3 Parameter Lock Icons

Adjacent to **Style**, **Strategy**, **Key/Scale**, and **Motif** are vector padlock icon buttons:

- **Unlocked (Open Padlock):** Parameters can be randomized when clicking **DICE**.
- **Locked (Closed Padlock, Glowing Cyan):** Freezes the parameter, protecting your key, scale, or favorite style from being modified when re-rolling patterns.

4.4 Generation Actions

- **NEW PAIR:** Generates a completely new correlated Call and Response phrase using current settings.
- **DICE:** Rerolls only the unlocked parameters and generates fresh phrases.
- **NOW vs NEXT PHRASE:** Sets regeneration timing. In *NEXT PHRASE* mode, pressing generate queues the transition seamlessly until the next 16-step bar boundary ("*QUEUED*").

5. Performance Knobs & 303 Synth

5.1 Performance Rotaries

- **DENSITY:** Dynamically thins or populates note events without altering the core downbeats.
- **GATE:** Adjusts global articulation from snappy staccato ticks (10%) to extended sustained ties (250%).
- **OCT JUMP:** Introduces dynamic ± 12 and ± 24 semitone octave leaps inspired by vintage 303 acid lines.
- **RESOLVE:** Directs phrase-ending notes toward harmonic tonic resolution or unresolved tension.

5.2 303 Preview Synthesizer

- **CUTOFF:** Adjusts low-pass filter cutoff frequency.
- **RESONANCE:** Amplifies resonant frequency peak for classic squelch.
- **DECAY:** Shapes the envelope decay release time.

6. Sequencer, Piano Roll & Split Views

Use the top view switcher to choose your preferred editing workspace:

6.1 Step Grid View (`[STEPS]`)

Features independent lanes for **CALL** and **RESPONSE**:

- **MIDI Channel Routing:** Assign each lane to distinct MIDI output channels (Channel 1–16) for driving multitimbral plugins or hardware.
- **Lane Actions:**
 - **MUTE**: Silences that lane independently.
 - **< / >**: Shifts the pattern horizontally left or right.
 - **-8ve / +8ve**: Transposes the entire lane up or down one octave.
 - **Inv**: Inverts scale degrees around the median pitch.
 - **Rev**: Reverses the sequence in time.
- **Step Editing:**
 - **Double-Click:** Adds or deletes a note at that step.
 - **Drag Horizontally:** Moves and swaps notes between steps left and right.
 - **Drag Vertically:** Changes scale degree / pitch.
 - **Right-Click:** Toggles **ACC** (Accent).
 - **Alt-Click:** Toggles **SLD** (Slide / Portamento).

6.2 Piano Roll View (`[PIANO ROLL]`)

Full interactive melody editor displaying pitch against time:

- **2D Note Manipulation:**
 - Click and drag a note **Up/Down** to transpose pitch with real-time audio auditioning.
 - Drag **Left/Right** to move notes across steps without scrambling neighboring columns.
 - Hovering note center displays 4-way move cursor ().
- **Double-Click:**
 - Double-clicking an empty cell adds a note conforming to the current scale.
 - Double-clicking an existing note deletes it.
- **Gate Resizing:** Hover over the right edge of any note (cursor changes to ) and drag horizontally to customize gate length.
- **Pitch Audition:** Click any piano key on the left keyboard to preview notes.
- **Articulations:**
 - Right-click note to toggle Accent (orange badge).
 - Alt-click note to toggle Slide (cyan slide line).
 - Shift-click note to delete.

6.3 Split View (`[SPLIT]`)

Simultaneously displays Step Grids on top with the Piano Roll below:

- **BOTH (Default)**: Stacked Call and Response step grids above the combined Piano Roll.
 - **CALL**: Call Step Grid on top with Call Piano Roll below.
 - **RESP**: Response Step Grid on top with Response Piano Roll below.
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7. Dynamic Scale & Key Transposition

Whenever you change the **Key**, **Scale**, or **Root Octave** dropdowns:

- Existing notes in both Call and Response are **automatically transposed** to conform to the new scale.
 - **Musical Function Preservation**: Preserves modal relationships (minor 3rd $\rightarrow$ major 3rd, minor 6th $\rightarrow$ major 6th, minor 7th $\rightarrow$ major 7th, minor 2nd $\rightarrow$ major 2nd, tritone $\rightarrow$ perfect 5th).
 - Transpositions reflect immediately in the **Piano Roll**, the **Step Grid**, the **Preview Synth**, and **Exported MIDI**.
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8. Mutation Engine & MIDI Export

8.1 Mutation Engine

Select from **9 surgical mutation targets**:

1. **Rhythm only**: Syncopates and shifts step timing while preserving pitch.
2. **Pitch only**: Modifies scale degrees while preserving rhythm.
3. **Articulation only**: Re-rolls gate lengths, accents, and slides.
4. **Ending only**: Rewrites the final cadence steps (steps 13–16).
5. **Motif only**: Reinvents the opening 2–4 note theme.
6. **Whole phrase**: Comprehensive variation of rhythm, pitch, and feel.
7. **Response only**: Mutates exclusively the response lane.
8. **Call only**: Mutates exclusively the call lane.
9. **Call + response linked**: Mutates call and procedurally re-derives response.

Amount options: **Tiny (1 step)**, **Small (2 steps)**, **Medium (3 steps)**, **Strong (5 steps)**, **Brutal (8 steps)**.

8.2 MIDI Drag & Drop

Click and drag the **DRAG MIDI** button directly into your DAW arrangement or desktop:

- **Export Modes**:
 - **Combined**: Exports Call followed sequentially by Response in a single MIDI clip.
 - **Call Only**: Exports only the Call pattern.
 - **Response Only**: Exports only the Response pattern.

- **Layered** : Exports Call on Channel 1 and Response on Channel 2 simultaneously.
 - **Whole Bars**: Pads the export to full musical bar lengths for seamless DAW looping.
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9. File Locations & Installation Directory

9.1 macOS

- **VST3 Plugins:**
 - System: `/Library/Audio/Plug-Ins/VST3/Call & Response MIDI.vst3`
 - User: `~/Library/Audio/Plug-Ins/VST3/Call & Response MIDI.vst3`
- **Audio Unit (AU) Plugins:**
 - System: `/Library/Audio/Plug-Ins/Components/Call & Response MIDI.component`
 - User: `~/Library/Audio/Plug-Ins/Components/Call & Response MIDI.component`
- **Standalone Application:**
 - `/Applications/Call & Response MIDI.app`
- **Documentation:**
 - `/Library/Audio/Documentation/Viral Samples/Call & Response MIDI/`
- **Installer:**
 - `Call & Response MIDI v1.0.5 Installer.pkg`

9.2 Windows

- **VST3 Plugins:**
 - `C:\Program Files\Common Files\VST3\Call & Response MIDI.vst3`
 - **Standalone Application:**
 - `C:\Program Files\Viral Samples\Call & Response MIDI\Call & Response MIDI.exe`
 - **Documentation:**
 - `C:\Program Files\Viral Samples\Call & Response MIDI\Documentation\`
 - **Installer:**
 - `Call & Response MIDI v1.0.5 Windows Setup.exe` (built via Inno Setup script `installer/windows/installer.iss` or `build_windows.bat`)
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